

Course: B. Tech.

Branch : E & T C

Semester : V

Subject Code & Name: BTETC503 - Analog Communication Engineering

Max Marks: 60

Date:11/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve / Answer the following. (ANY TWO)		
A)	Define Analog Communication and Explain the elements of communication system	CO 01	6
B)	Why modulation is necessary in communication, and what purpose does it serve?	CO 01	6
C)	Describe the concept of multiplexing and its two main techniques: Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM).	CO 01	6
Q.2	Solve / Answer the following. (ANY TWO)		
A)	Explain DSBFC Technique in detail with the help of (i) frequency spectrum (ii) Time domain representation (iii) power relation with carrier wave	CO 02	6
B)	A transmitter radiates 10 KW power with the carrier unmodulated, and 10.5 kW when the carrier is modulated by one sinusoidal signal. Calculate the modulation index. If another modulating signal corresponding to 30% modulation is transmitted simultaneously determine the total radiated power	CO 02	6
C)	Provide the mathematical analysis and expression for Amplitude Modulation.	CO 03	6
Q. 3	Solve / Answer the following. (ANY TWO)		
A)	What is Angle modulation? Derive the mathematical expression for FM with frequency spectrum .	CO 05	6
B)	Discuss the concepts of pre-emphasis and de-emphasis in FM systems and their role in improving signal quality.	CO 05	6
C)	It is required to provide a maximum deviation of 75 kHz for the 88-MHz carrier frequency of a VHP FM. transmitter. A FET is used as a capacitive reactance modulator, and the linear portion of its g-curve lies from 320 μ S (at which V= -2V) to 830 μ S (at which V= -0.5V). Assuming that R is one-tenth of Xc calculate (a) rms value of required modulating voltage (b) The value of the fixed capacitance and inductance of the oscillator tuned circuit across which the reactance modulator is connected	CO 05	6

Q.4 Solve / Answer the following. (ANY TWO)

- A) Explain the key performance characteristics of receivers: Sensitivity, Selectivity, and Fidelity. CO 07 6
- B) Describe the Tuned Radio Frequency (TRF) receiver architecture and its advantages and disadvantages. CO 04 6
- C) Explain the operation of a Phase Discriminator in FM demodulation, and highlight its advantages and disadvantages. CO 04 6

Q.5 Solve / Answer the following. (ANY TWO)

- A) Explain the operation of an Envelope Detector in AM demodulation, and discuss its advantages and limitations. <https://www.batuonline.com> CO 04 6
- B) Define Noise Figure , Noise Factor, and explain their relationship in evaluating the noise performance of electronic devices. CO 04, 06 6
- C) An AF. signal $20 \sin(2\pi * 500t)$ is used to amplitude modulate a carrier of $50 \sin(2\pi * 10^5 * t)$ Calculate: CO 04, 06 6
- I. Modulation index
 - II. Sideband frequencies
 - III. Amplitude of each sideband frequency
 - IV. Bandwidth required
 - V. Total power delivered into a load of 600Ω

***** End *****